Identifying the Risk in Lie Detection for Assessing Guilty and Innocent Subjects for Healthcare Applications

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Abstract *Objective*: This study aims to critically assess the implications and potential hazards of employing deep learning algorithms for the identification of innocence and guilt through lie detection methodologies within healthcare environments. *Methods*: The research centres on exploring Electroencephalography (EEG) as a significant technique for lie detection, particularly via the Concealed Information Test (CIT). CIT, a forensic approach, is deployed to identify hidden or deceptive information by monitoring physiological reactions to specific queries containing sensitive information pertinent to a case. This study highlights the importance of band-pass filters in lie detection, which enable the isolation of particular EEG signal frequencies that may reflect cognitive processes associated with deception. *Findings*: An exhaustive examination of various machine learning algorithms, including Support Vector Machines, Linear Support Vector Classification, Multilayer Perceptron, alongside deep learning frameworks like Long Short-Term Memory (LSTM) and Deep Neural Networks (DNN), was conducted, incorporating sophisticated feature extraction techniques. The investigation reveals critical considerations and potential risks linked with the application of these technologies for distinguishing between guilty and innocent subjects in healthcare applications, emphasizing the need for careful and ethical use of such advanced detection methods. *Novelty*: The outcomes of our study reveal that the Multilayer Perceptron (MLP) attained an impressive accuracy rate of 98.36%, while the Deep Neural Network (DNN) demonstrated a high accuracy of 98.28%.

Keywords Electroencephalography · Lie detection · Concealed information test · Guilty · Innocent

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1 Introduction

Brain fingerprinting, an innovative and developing field, seeks to detect specific brain activity patterns linked to an individual's recognition or awareness of particular information or incidents. This method is increasingly being applied to ascertain if a person holds detailed knowledge related to a crime, event, or information crucial to an investigation. By measuring and analyzing brainwave responses, especially focusing on the P300 wave—a marker for recognizing familiar stimuli—brain fingerprinting serves multiple purposes. It is utilized in criminal justice, counterterrorism, and intelligence operations to verify a subject's acquaintance with crime-specific information or sensitive data [1].

The utility of brain fingerprinting extends to verifying personal involvement in criminal activities or possession of confidential knowledge, primarily when conventional evidence-gathering techniques fall short. Lie detection, another significant aspect, strives to identify when individuals intentionally mislead or conceal the truth during interrogations. Traditional methods, such as polygraph tests, analyze physiological responses like heart rate, blood pressure, and skin conductivity to infer deception. These practices are prevalent in legal, security, and employment contexts to evaluate the reliability of statements made by witnesses, suspects, or job applicants [2].

Employing EEG technology enhances our understanding of the brain's reaction to deceptive scenarios. The Polygraph test, based on physiological responses, and EEG-based Concealed Information Test (CIT), which presents different stimuli (probe, irrelevant, and target) to subjects, are critical in this domain. The CIT aims to identify a guilty individual's recognition of specific information by observing the P300 response to probe stimuli, distinguishing it from irrelevant stimuli perceived by innocent individuals. This differentiation is crucial in identifying deceptive versus innocent behavior [3].

The drive towards creating lie detection systems using deep learning stems from the need for more precise, unbiased, and efficient methods to separate truth from deceit. Deep learning offers a promising avenue for enhancing the accuracy and reliability of lie detection by analyzing complex patterns of physiological, behavioral, and verbal cues. This approach aims to minimize subjective interpretation, focusing on developing ethical, privacy-aware systems that ensure consent and uphold individual rights. The overarching goal is to refine deception detection technology, contributing to its application in security, law enforcement, and legal settings, thereby ensuring a higher integrity level in truth verification processes. In the realm of applying EEG (Electroencephalography) for distinguishing between truth and deceit in healthcare contexts, particularly for identifying guilt or innocence, an in-depth exploration of EEG signal processing and the methodologies for extracting features is essential for a thorough comprehension of these approaches and their underlying justifications [4, 5].

The process of EEG signal analysis encompasses several key stages aimed at improving the clarity of the EEG signals and extracting pivotal information for subsequent examination. Initial stages typically involve preprocessing, which tackles filtering, the elimination of artifacts, and the segmentation of signals. The application of band-pass filters is crucial for discarding irrelevant noise and honing in on specific frequency ranges associated with cognitive processes linked to lying, including theta, alpha, and beta waves. The removal of artifacts is vital for eradicating interferences from external sources, such as electrical appliances or physical movements, thereby ensuring the EEG data's integrity for precise analysis. After preprocessing, the focus shifts to extracting features from the EEG signals that signify deceptive behaviors. Techniques for feature extraction vary, encompassing aspects from the time domain (like signal amplitude and variability), frequency domain (including power spectral density), and time–frequency domain (such as coefficients obtained through wavelet transforms). These techniques are adept at capturing crucial aspects of the brain's reaction to stimuli, highlighting disparities in neural activity between honest and dishonest responses [6].

The emphasis on these particular methods of signal processing and feature extraction is driven by their capacity to shed light on the neural underpinnings of dishonesty. The P300 wave, for instance, is a critical marker, identifiable as a positive shift in the EEG signal around 300 ms following the presentation of a stimulus, particularly noticeable when the subject recognizes related stimuli. This phenomenon serves as a crucial indicator for differentiating between truthful and deceptive replies. Through the analysis of such features, the detection of hidden knowledge becomes feasible, thus facilitating lie detection processes.

In healthcare applications where the accuracy and ethical deployment of EEG-based lie detection are of utmost importance, these signal processing and feature extraction methodologies present a non-intrusive, objective, and scientifically backed means to judge an individual's veracity or deception. Advancing these techniques and deepening the understanding of their foundational principles will enhance the effectiveness and ethical use of lie detection within healthcare settings, ensuring it plays a vital role in accurately assessing individuals' guilt or innocence.

In healthcare, the motivation behind EEG lie detection research lies in the quest for precise, non-invasive methods to detect deception, which could have significant implications for patient care. EEG's ability to capture real-time brain activity with high precision offers potential insights into the cognitive processes underlying patient communication and compliance. Ethical considerations drive the development of standardized, objective approaches, ensuring patient autonomy and trust. This research holds promise for practical applications in healthcare settings, including assessing patient-reported symptoms, evaluating treatment adherence, and detecting potential maladaptive behaviors. By leveraging EEG lie detection in healthcare, clinicians can enhance patient care by fostering open and honest communication while respecting patient confidentiality.

2 Materials and Methods

Empirical Mode Decomposition (EMD) addresses the limitations of traditional polygraph tests, while EEG-based lie detectors focus on covert behavior through ERP EEG components. This study employs EMD for EEG feature extraction, followed by classification using Support Vector Machines (SVM), Quadratic Discriminant Analysis (QDA), K-Nearest Neighbours (KNN), and decision trees. SVM often outperforms in distinguishing guilt from innocence [1]. The Deceit Identification Test utilizes EEG, particularly P300 signals, achieving a commendable 95% classification accuracy with symlet Wavelet Packet Transform (WPT) for feature extraction [2]. Employing a k-nearest neighbour classifier, it attains an impressive 96.7% classification accuracy, notably in mobility analysis [3]. Using WPT and LDA, this study reaches up to 91.67% accuracy [4]. Introducing a 3-stage hybrid classification method, it outperforms existing methods with an 83.1% accuracy rate, demonstrating effectiveness in concealed information detection [5]. In analysing the DEAP dataset, the model achieves recognition accuracies of 89.49% for valence and 92.86% for arousal prediction, while applying it to the SEED dataset results in a remarkable 96.77% recognition accuracy. The proposed network, combining Convolutional Neural Networks (CNN), Stacked Auto encoders (SAE), and Deep Neural Networks (DNN), offers remarkable efficiency and faster convergence, promising enhanced accuracy and efficiency in dataset analysis and prediction [6]. Despite various EEG classification methods, deep belief networks remain underexplored in lie detection. This study utilizes a deep learning strategy incorporating a restricted Boltzmann machine alongside wavelet analysis to capture EEG characteristics in both the time and frequency domains. A four-layer deep belief network, constructed by stacking RBMs, successfully classifies EEG data through softmax regression. An experiment involving EEG data from a "Concealed Information Test" captures brain responses to relevant and irrelevant images [7]. Non-invasive LSTM and bidirectional LSTM models are designed to extract subject-specific mental task-related features from EEG data. Training involves the average combined Power Spectrum Density (PSD) values from all subjects except one. The deep neural network (DNN)-based model achieves an accuracy of 77.62%, demonstrating its capability to effectively capture task-specific EEG features, making it a promising approach for mental task analysis [8]. The lie detection model, utilizing the Fast Fourier Transform and the Minimize Entropy Principle Approach (MEPA), attains an impressive accuracy rate of 89.5%. Additionally, a mobile prototype has been developed, integrating standard brainwave measurement tools with mobile devices, enabling real-time and precise polygraphy. This advancement marks a significant step toward efficient and accurate lie detection in real-world applications [9]. An SVM-based approach achieves an impressive 96.67% classification accuracy for EEG signals. A validation test using unclassified EEG data further assesses the algorithm's performance [10]. Numerous features encompassing morphological-temporal, frequency, and wavelength characteristics are extracted from three signal types. The classification approach utilizes

K-nearest neighbours (KNN) and incorporates validation techniques. An innovative application of the cuckoo optimization algorithm (COA) as a feature selection tool identifies the most relevant feature subset. The study demonstrates the capability to accurately categorize participants into two groups with an impressive accuracy rate of 93.3%, particularly highlighting improved classification performance when combining features from respiration, reaction signal, and EEG [11]. The study employs linear discriminant analysis (LDA) for distinguishing between positive and negative samples within sensor-acquired signals, achieving an impressive 85% accuracy in correctly identifying deception in both guilty and innocent subjects. The entire system is implemented on a Field-Programmable Gate Array (FPGA), optimizing efficiency [12]. Two multi-person data fusion techniques, parallel and serial data fusion, integrate EEG data from multiple individuals responding to the same task instructions. The fused data serves as input for a Convolutional Neural Network (CNN) designed for single-trial P300 classification, achieving an accuracy of 83.03% [13]. A deep learning approach employs a Convolutional Neural Network (CNN) for automated truth detection from EEG signals, achieving an impressive accuracy of up to 84.44% in discerning truth from lies. This non-invasive, efficient, and robust approach with minimal time complexity is well-suited for real-time applications [14]. The proposed method is validated using a high-density EEG dataset containing 128 channels. It achieves a 93.33% accuracy using K-nearest neighbours (KNN) for the classification of two cognitive tasks, demonstrating promising results for cognitive task analysis [15]. ERP components P200 and N200 distinguished instructed truth-telling. N300 separated instructed lies from spontaneous conditions. Late positive potential indicated spontaneous lying, correlated with risk-taking tendencies. ERS alpha and low beta discriminated conditions effectively. Single-trial decoding achieved robust performance [16]. The V-TAM model employs distinct pathways for extracting local and global features in spatial and temporal domains. It introduces a novel Visual-Temporal Attention Model (V-TAM) to improve EEG segmentation precision, achieving 98.5% accuracy in lie detection tasks, outperforming three recent models [17]. Detecting lies in the digital age remains challenging for law enforcement and researchers. A novel approach, using Arduino technology, leverages physiological changes to identify deception [18]. An improved Recurrent Neural Network (RNN), termed Enhanced RNN (ERNN), incorporates Explainable AI features via LSTM architecture. Fuzzy logic optimizes hyper parameters. Trained on interview audio data, ERNN achieves 97.3% accuracy in voice stress analysis, offering explainable insights into stress patterns [19].

2.1 Major Contributions

Numerous research endeavours focus on improving lie detection accuracy through novel EEG-based classification algorithms or enhancements to existing ones, potentially yielding higher classification rates. Studies shedding light on cognitive and physiological aspects of deception enrich our comprehension of human behavior, often revealing discernible patterns in EEG data indicative of deceit. Investigations frequently delve into optimizing feature selection and extraction methodologies to pinpoint informative EEG markers distinguishing truthful from deceptive responses, thereby advancing the field's efficacy. Various classifiers, such as SVM, Linear SVM, MLP, LSTM, and DNN, are applied to EEG data from diverse subjects to distinguish between truthful and deceptive responses. MLP and DNN demonstrate superior performance, with DNNs automatically extracting intricate features crucial for lie detection from complex EEG data.

Figure 1 shows flowchart for Classification of EEG data into Guilty and Innocent object:

1. **EEG Signal**: It starts with the EEG signal acquisition, where electrical activity from the brain is captured.
2. **BrainVision Recorder and Analyser**: This hardware/software system records the brain's electrical signals and prepares them for analysis.

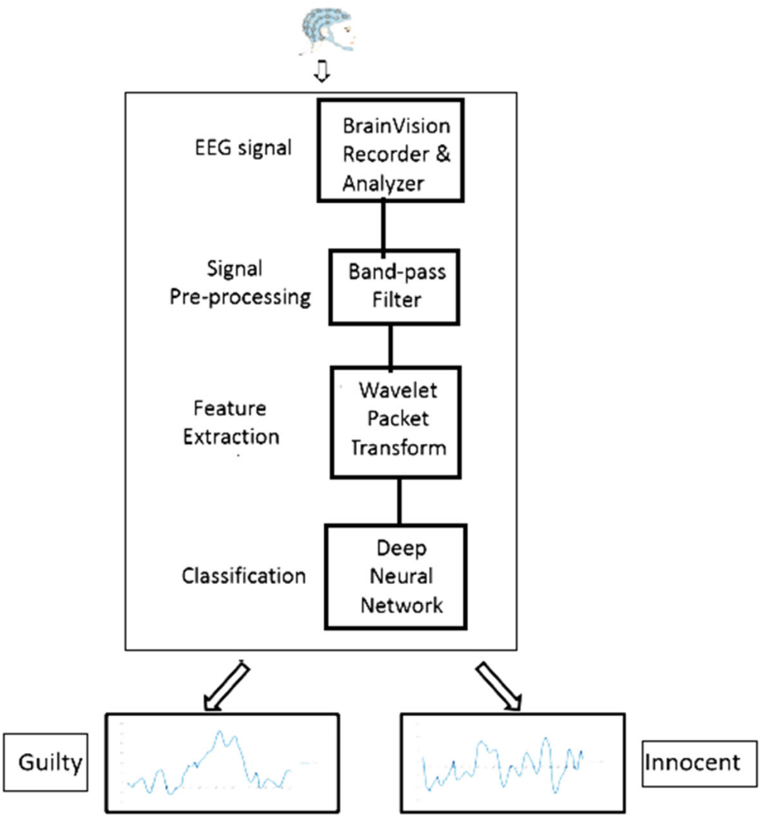

Fig. 1 Classification of EEG data into guilty and innocent object

3. **Signal Pre-processing**: The raw EEG data undergoes initial processing, which may include noise reduction and signal enhancement. This is often performed using band-pass filters to isolate the frequency ranges of interest.
4. **Feature Extraction**: After pre-processing, the next step is to extract meaningful features from the EEG data. This is typically done using methods like Wavelet Packet Transform, which breaks down the signal into wavelets or small waves, capturing both frequency and location information.
5. **Classification**: The extracted features are then fed into a Deep Neural Network (DNN), an advanced machine learning model that identifies patterns and learns to classify the input data as 'Guilty' or 'Innocent'.
6. **Outcome**: The final output of the process is a classification based on the EEG signal's characteristics, represented here by two different EEG waveforms, each labelled as 'Guilty' or 'Innocent'. These labels suggest that the system is used for some form of forensic or lie detection analysis, where EEG patterns are used to infer the subject's truthfulness or culpability.

Figure 2 shows that **four lobes** of the brain are the frontal, parietal, temporal, and occipital lobes

1. **Frontal Lobe**: Positioned at the front of the brain, the frontal lobe is associated with higher-order cognitive tasks such as problem-solving, planning, decision-making, reasoning, and voluntary muscle control. It also houses the motor cortex, responsible for controlling voluntary muscle movements.
2. **Parietal Lobe**: Located near the top and back of the brain, the parietal lobe plays a role in processing sensory information related to touch, temperature, and spatial awareness. Within this lobe, the somatosensory cortex interprets tactile sensations.
3. **Temporal Lobe**: Found on the sides of the brain, the temporal lobe is responsible for functions like auditory processing, memory formation, comprehension of

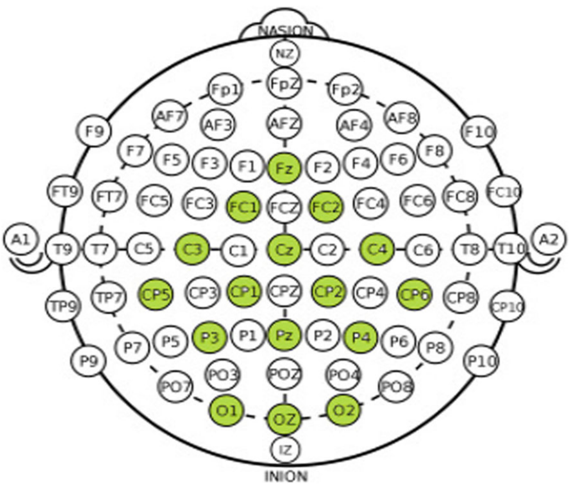

Fig. 2 Different brain lobes and their location

language, and recognition of faces and objects. The hippocampus, crucial for memory, resides within the temporal lobe.

4. **Occipital Lobe**: Situated at the back of the brain, the occipital lobe is primarily dedicated to processing visual information received from the eyes. It is essential for visual perception, object recognition, and the interpretation of visual stimuli.

The performance metrics used in research as follows:

(1) Accuracy: Accuracy measures the proportion of correctly classified instances out of all instances in the dataset.

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} = \frac{TP + TN}{TP + TN + FP + FN}$$

(2) Recall: Recall measures the proportion of true positive instances correctly identified by the model out of all actual positive instances.

$$\text{Recall} = \frac{TP}{TP + FN}$$

(3) AUC-ROC: The AUC-ROC metric gauges a model's capacity to differentiate between positive and negative classes. It quantifies the area under the curve of the receiver operating characteristic (ROC) curve, depicting the relationship between the true positive rate (recall) and the false positive rate (FPR) across different threshold configurations.

True Positive (TP)	**False Positive (FP)**
False Negative (FN)	**True Negative (TN)**

$$TPR = \frac{TP}{TP + FN}$$

$$FPR = \frac{FP}{FP + TN}$$

where

- TP (True Positives): Refers to the count of positive instances accurately identified as positive.
- TN (True Negatives): Denotes the quantity of negative instances correctly identified as negative
- FP (False Positives): Represents the tally of negative instances mistakenly classified as positive.
- FN (False Negatives): Indicates the number of positive instances erroneously classified as negative.

3 Results and Discussion

The dataset consists of 3 objects such as neutral, innocent and guilty.

Figure 3 shows categorization of dataset in term of neutral, innocent and guilty.

Figure 4 shows t-SNE visualization of subjects such as Neutral, Innocent and Guilty. t-SNE, a dimensionality reduction method, visualizes complex lie detection data in lower dimensions while maintaining local relationships. It aids in understanding deceptive patterns, crucial for enhancing accuracy in lie detection systems.

Figure 5 shows values from the datasets. From above dataset two types of features are categorized such as significant and Nonsignificant features based on p_values from datasets.

If p_value < 0.05 then significant features
Else
Non-significant features

Figure 6 shows categorization of significant and Nonsignificant features. Figure 6 depicts neutral, innocent and guilty objects in terms of significant and Nonsignificant features.

Comparison with Previous Machine Learning Classifiers

Following Table 1 shows different ML classifiers used in the previous research and accuracy is specified. Figure 7 shows comparison in graph that depicts SMO and SM-CRM provides 97.66% accuracy.

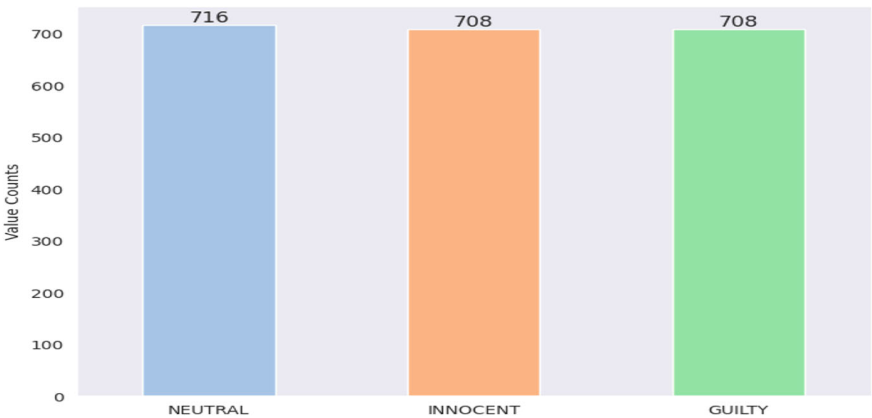

Fig. 3 Categorization of subjects

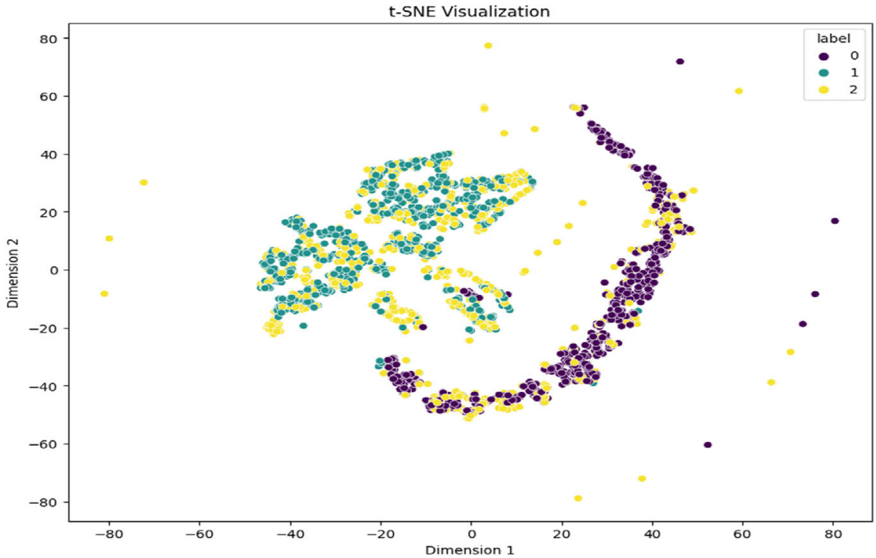

Fig. 4 t-SNE visualization of subjects such as neutral, innocent and guilty

	# mean_0_a	mean_1_a	mean_2_a	mean_3_a	mean_4_a	mean_d_0_a	mean_d_1_a	mean_d_2_a	mean_d_3_a	mean_d_4_a	...	fft_741_b	fft_742_b	fft_743_b	fft_744_b	fft_745_b	fft_746_
0	4.62	30.3	-356.0	15.6	26.3	1.070	0.411	-15.70	2.06	3.15	...	23.5	20.3	20.3	23.5	-215.0	280.0
1	28.80	33.1	32.0	25.8	22.8	6.550	1.680	2.88	3.83	-4.82	...	-23.3	-21.8	-21.8	-23.3	182.0	2.5
2	8.90	29.4	-416.0	16.7	23.7	79.900	3.360	90.20	89.90	2.03	...	462.0	-233.0	-233.0	462.0	-267.0	281.0
3	14.90	31.6	-143.0	19.8	24.3	-0.584	-0.284	8.82	2.30	-1.97	...	299.0	-243.0	-243.0	299.0	132.0	-12.4
4	28.30	31.3	45.2	27.3	24.5	34.800	-5.790	3.06	41.40	5.52	...	12.0	38.1	38.1	12.0	119.0	-17.6

5 rows × 2549 columns

Fig. 5 Values from datasets

Comparison with Previous Deep Learning Classifiers

Following Table 2 shows different DL classifiers used in the previous research and accuracy is specified. Figure 8 shows comparison in graph that depicts CNN, SAE and DNN provides 96.77% accuracy.

Results with Proposed Algorithm

Following Table 3 shows different ML and DL classifiers used in the previous research and accuracy is specified. MLP provides 98.36% better accuracy. DNN provides 98.28% better accuracy. Figure 9 shows comparison in graph that depicts MLP and DNN provide better accuracy as compare to other ML and DL classifiers respectively.

Various classifiers, including SVM, Linear SVM, MLP, LSTM, and DNN (Deep Neural Network), are utilized to process EEG data collected from individual subjects.

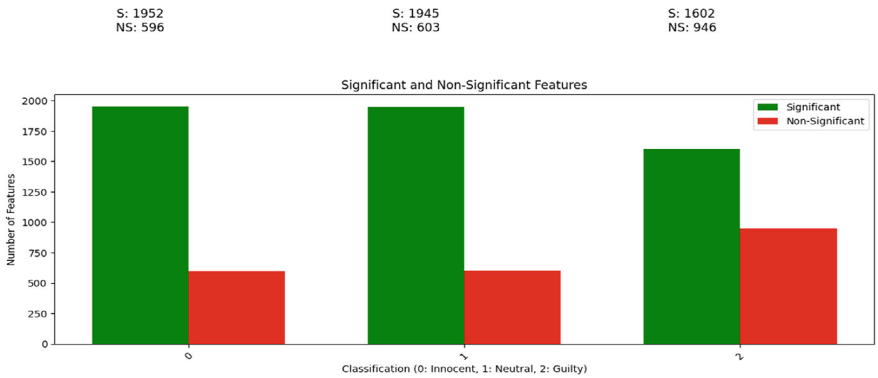

Fig. 6 Distribution of significant and non-significant features

Table 1 Comparison with previous ML classifiers

Sr. No.	Methodology	Accuracy (%)
1	SVM	81.71
2	Deep belief network	81.03
3	KNN (K nearest neighbor)	96.7
4	Linear discriminant analysis (LDA)	91.67
5	Multistage hybrid classifier	83.1
6	Spider monkey optimization (SMO) algorithm and SM-candidate rule miner (SM-CRM)	97.66

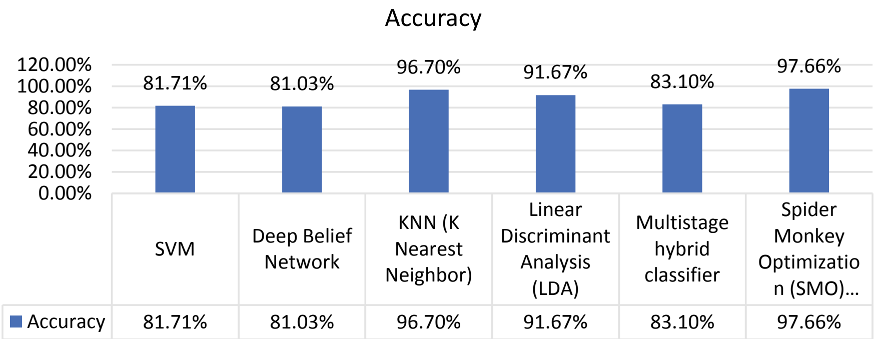

Fig. 7 Comparison with previous ML classifiers

A series of innovative experiments are conducted involving diverse participants to discern between truthful and deceptive responses. Analysis of the outcomes reveals that MLP and DNN exhibit superior performance compared to other classifiers when

Table 2 Comparison with previous DL classifiers

Sr. No.	Methodology	Accuracy (%)
1	DNN	76.67
2	CNN, SAE and DNN	96.77

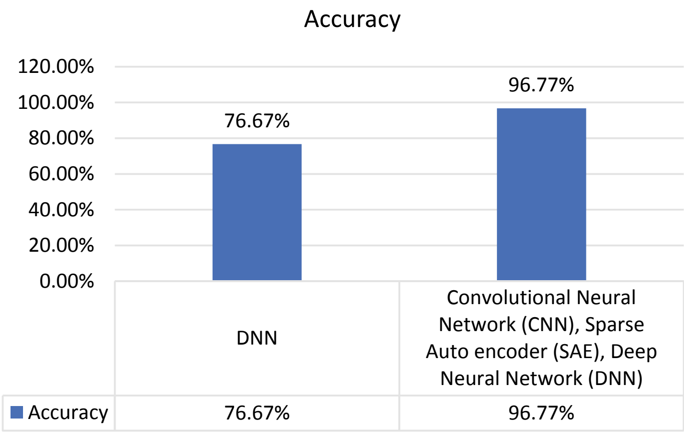

Fig. 8 Comparison with previous DL classifiers

Table 3 Results with proposed algorithm

Sr. No.	Methodology	Accuracy
1	SVM	96.95
2	Linear SVC	96.72
3	**MLP**	**98.36**
4	LSTM	96.87
5	**DNN**	**98.28**

applied to the recorded EEG dataset. DNNs are capable of automatically extracting and learning high-level features from raw EEG data, a process that is significantly more complex for traditional machine learning models. This automatic feature extraction is crucial for lie detection, where the relevant signals are often deeply embedded in the EEG data and may be indiscernible to simpler models. DNNs can uncover intricate patterns within the data that are indicative of deceptive behavior, which other classifiers might overlook. EEG data is inherently high-dimensional, with complex relationships across different frequencies, channels, and time points. DNNs excel in managing this complexity through their deep architecture, which can process and integrate information across multiple layers. This capability allows DNNs to better understand the spatial and temporal dynamics of the brain's response to deceptive stimuli, leading to more accurate lie detection.

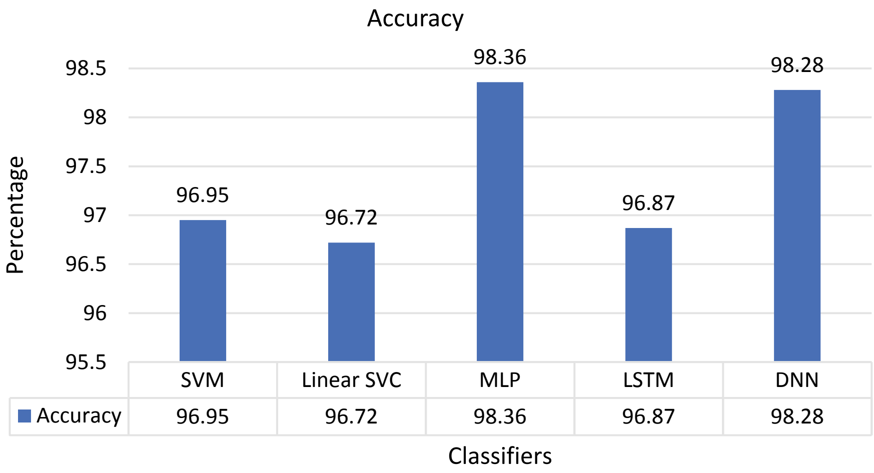

Fig. 9 Comparison with proposed ML and DL classifiers

A DNN architecture for lie detection through EEG analysis is a sophisticated configuration that leverages convolutional and fully connected layers, complemented by effective activation functions and optimization algorithms. This setup is engineered to process the complex, high-dimensional EEG data, learning to differentiate between deceptive and truthful responses accurately. The choice of layers, functions, and algorithms is critical to the model's performance, dictating its ability to learn from the data without overfitting and its efficiency in generalizing from training to unseen data (Fig. 10).

DNNs leverage their advanced feature extraction, ability to handle complex data structures, robustness to individual variability, superior generalization capabilities, and flexibility in integrating multimodal data to offer unparalleled performance in EEG-based lie detection. These attributes make DNNs particularly effective in distinguishing between truthful and deceptive responses, outperforming other classifiers in the nuanced task of interpreting EEG data for lie detection purposes. The integration of Deep Neural Networks (DNN) into the realm of lie detection, particularly when using EEG data, represents a significant advancement over traditional methodologies. This comparison outlines the benefits and drawbacks of the proposed DNN approach compared to existing lie detection techniques, such as the Polygraph test and basic machine learning algorithms. In the domain of lie detection, these machine learning and deep learning techniques are typically trained on annotated datasets containing both deceptive and truthful responses. This training enables them to learn and recognize patterns and features associated with deception. Once trained, they can be deployed on new data to classify responses, aiding in the identification of potential deception indicators. The deployment of EEG-based lie detection technologies in real-world scenarios raises significant ethical considerations and potential societal impacts that warrant careful examination. As these technologies advance and become

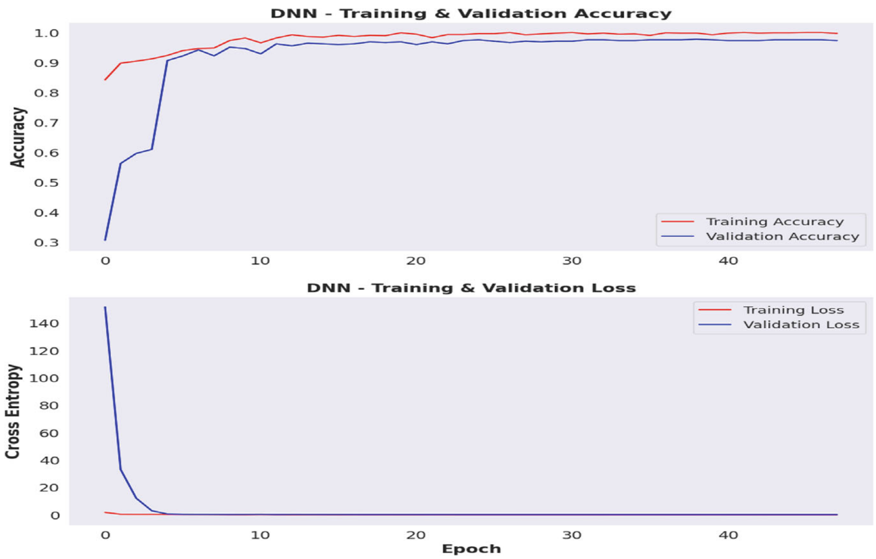

Fig. 10 Training and validation accuracy using DNN

more accessible, their integration into various sectors, including law enforcement, legal systems, and employment screening, must be approached with a keen awareness of the ethical implications and the broader consequences for society.

4 Conclusion

A key task is to differentiate between innocent and guilty individuals using EEG data in lie detection. The study aimed to address the limitations of existing systems and provide a more accurate, reliable, and practical approach to brain fingerprinting and lie detection. It has contributed to the understanding of brainwave patterns, deception detection, and the intersection of neuroscience and forensic sciences. Deception Detection Test, another forensic tool, identifies concealed information by monitoring reactions to critical questions, focusing on detecting hidden knowledge rather than veracity, and is widely utilized in various investigative and security scenarios. This study, a comprehensive exploration of machine learning techniques, including Support Vector Machine, Linear SVC, Multilayer Perceptron, and deep learning techniques, including LSTM, Deep Neural Network (DNN) and advanced feature extraction methodologies, has been undertaken. According to results, MLP 98.36% achieved accuracy and DNN 98.28% achieved accuracy. Future directions in EEG-based lie detection involve refining signal processing, integrating multiple modalities, developing real-time systems, considering individual variations, addressing ethical concerns, and validating applications across diverse contexts. This aims to enhance reliability and practicality in security, psychology, and human–computer interaction domains.

Declarations "The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper."

Author Contributions Example:

Tanmayi Nagale: Conceptualization, Methodology, Software Development; Data Collection, Drafting of the Original Manuscript. Visualization and Analysis.

Dr. Anand Khandare: Methodology, writing, reviewing, and editing.

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